



SCIENTIFIC ANALYSIS GROUP

Defense Research & Development Organisation

**QSAFire: Design and Performance Evaluation of
Satellite-Assisted SNS-TF-QKD over the Delhi-Mumbai
Quantum Communication Link**

Internship Report

Submitted in partial fulfilment of the requirements
for the award of the degree of

Bachelor of Technology

in Department of Electronics and Communication Engineering

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June- August 2026

Certificate

This is to certify that the internship report titled “**QSAFire: Design and Performance Evaluation of Satellite-Assisted SNS-TF-QKD over the Delhi-Mumbai Quantum Communication Link**” is a bona fide record of the work carried out by **Anand Kumar**, a student of Bachelor of Technology (Department of Electronics and Communication Engineering), Bharati Vidyapeeths’ College of Engineering, during an internship of duration June- August 2026 at the Scientific Analysis Group, Defence Research and Development Organisation (DRDO), Metcalfe House, Delhi.

The work reported here was carried out under my supervision and, to the best of my knowledge, has not been submitted elsewhere for the award of any other degree or diploma. The report is submitted for evaluation as part of the internship requirements of the student’s academic programme.

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Acknowledgement

I would like to thank Mr. Rajesh Kumar, Scientist ‘E’, Scientific Analysis Group, DRDO, Metcalfe House, for agreeing to supervise this internship and for the direction he gave at points where the work could easily have gone off track. The scope of the three tasks reported here, an OptiSystem prototype, a CSV preprocessing pipeline, and an independent satellite-link simulation study, was shaped substantially by conversations with him, and several of the modelling choices described in later chapters (in particular the decision to keep the physical-layer OptiSystem model and the protocol-layer Python analysis strictly separate) came directly out of that guidance.

I am also grateful to the Scientific Analysis Group at DRDO for the working environment and access to the software needed to carry out this work, and to the scientists and staff at Metcalfe House who answered questions along the way, even when those questions were basic ones.

Finally, I thank myself for my support during the weeks this internship took, including the days when a simulation refused to converge and the writing of this report ran later than planned.

Anand Kumar

Abstract

This report documents three technical tasks carried out during an internship at the Scientific Analysis Group, DRDO, all concerned in one way or another with Sending-or-Not-Sending Twin-Field Quantum Key Distribution (SNS-TF-QKD), but which were executed as three separate and largely independent pieces of work rather than as stages of a single pipeline. They are presented in this report exactly as they were carried out, without artificially merging them into one workflow.

The first task is an OptiSystem implementation of an SNS-TF-QKD optical link, built around three stations, Alice, Bob, and an untrusted measurement station Charlie, connected by fibre spans and driven at the physical layer by CW lasers, Mach-Zehnder and intensity modulators, and single-photon detectors, with the protocol-level decision logic (window selection, send/not-send choice, decoy-intensity selection, phase randomisation) implemented through MATLAB co-simulation blocks embedded inside the OptiSystem schematic. This is a proof-of-concept implementation at laboratory scale (40 km fibre spans, a 10 Gbit/s symbol clock, a 1024-bit sequence per run) and does not include any satellite or orbital modelling of any kind. The second task is a Python preprocessing pipeline written specifically to handle the CSV files that the OptiSystem model exports. Its job is to load the raw per-window logs produced by Alice, Bob, and Charlie's detectors, align them by bit index, clean up formatting inconsistencies in the exported data, and compute the basic sifting, gain, and error-rate quantities needed before any secure-key-rate estimate can be attempted. This pipeline operates only on OptiSystem-generated data and is not connected in any way to the satellite simulation framework described next.

The third task, entirely separate from the first two, is a Python satellite-link performance evaluation framework built from first principles: orbital geometry (500 km sun-synchronous, inclination solved from J2 perturbation theory), elevation-dependent slant range, atmospheric and Rytov-variance turbulence loss, pointing loss, and a fully asymmetric SNS-TF-QKD secure-key-rate calculation, applied to six Indian ground sites and four candidate links. The primary link studied, Delhi-Mumbai, returns a simulated secure key rate of 0.0947 kbit/s over 1149.4 km, roughly two orders of magnitude below Micius's own flagship figures but consistent in order of magnitude with real, non-ideal portable ground-station demonstrations reported in the literature. The framework's orbital and loss-budget formulas are cross-checked directly against Micius's published in-orbit performance (Liao et al., 2017) to within about 1 to 3 percent, and the analysis identifies night-sky background light, rather than atmospheric turbulence, as the domi-

nant factor limiting the Delhi-Mumbai link.

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List of Abbreviations

Abbreviation	Expansion
AOD	Aerosol Optical Depth
BB84	Bennett-Brassard 1984 (quantum key distribution protocol)
CSV	Comma-Separated Values
CW	Continuous Wave
DRDO	Defence Research and Development Organisation
FSO	Free-Space Optical
IM	Intensity Modulator
ITU-T	International Telecommunication Union, Telecommunication Standardization Sector
LOS	Line of Sight
MDI-QKD	Measurement-Device-Independent Quantum Key Distribution
MZM	Mach-Zehnder Modulator
NRZ	Non-Return-to-Zero
PM	Phase Modulator
PNS	Photon-Number-Splitting (attack)
PRBS	Pseudo-Random Bit Sequence
QBER	Quantum Bit Error Rate
QKD	Quantum Key Distribution
QSAFire	Quantum Satellite Analysis Framework for Integrated Re- search and Evaluation
SAG	Scientific Analysis Group
SKC	Secret-Key Capacity
SKR	Secure Key Rate
SMF	Single-Mode Fibre
SNS	Sending-or-Not-Sending
SNS-TF-QKD	Sending-or-Not-Sending Twin-Field Quantum Key Distribu- tion
SPD	Single-Photon Detector
TF-QKD	Twin-Field Quantum Key Distribution
WCP	Weak Coherent Pulse

List of Symbols

Symbol	Meaning
η	Channel transmittance
α	Fibre attenuation coefficient (dB/km)
L	Link distance (km)
ε	Sending probability in the SNS Z-basis
μ_1, μ_2	Signal and decoy mean photon numbers
k	Optical wavenumber, $k = 2\pi/\lambda$
C_n^2	Refractive-index structure parameter
σ_R^2	Rytov variance (scintillation index)
r_0	Fried parameter (atmospheric coherence length)
τ	Atmospheric optical depth
ζ	Zenith angle
e_a	Misalignment error parameter
QBER	Quantum bit error rate (fraction)
R	Secure key rate (bit/s or kbit/s)

Chapter 1

Introduction

1.1 Background and Motivation

Quantum key distribution (QKD) lets two parties agree on a shared secret key whose security rests on the physics of quantum measurement rather than on a computational hardness assumption [1]. This property is attractive for defence and government communication links, but standard point-to-point QKD is bound by a repeaterless rate-distance limit: the achievable key rate falls off exponentially with channel loss, which for fibre means an exponential fall-off with distance. Twin-Field QKD (TF-QKD) breaks this limit without a quantum repeater by having both parties send phase-coherent pulses to a central, untrusted measurement station and using the interference outcome, rather than a direct transmission, to generate correlated bits; this changes the key-rate scaling from linear to square-root in the overall channel transmittance [2]. The Sending-or-Not-Sending (SNS) variant of TF-QKD, used throughout this internship, removes a specific security gap in the original TF-QKD proposal by having each party make a private send/not-send decision for every key-generating time window instead of publicly disclosing phase information for those windows [3]. SNS-TF-QKD also tolerates a much larger single-photon misalignment error than the original scheme, which matters directly for any free-space or satellite link, where interference visibility is harder to maintain than in fibre.

This background is stated briefly here because it recurs, in different forms, across all three tasks reported in this document: Task 1 builds a physical-layer SNS-TF-QKD link in OptiSystem, Task 2 processes the CSV output of that link, and Task 3, the QSAFire framework, evaluates whether an SNS-TF-QKD link of this kind is plausible over a satellite downlink to Indian ground stations. None of the three tasks requires the reader to be re-taught QKD theory from first principles, and this report does not attempt to; the relevant protocol detail is introduced again, briefly, at the start of each task-specific chapter, only where it affects an implementation choice.

1.2 Scope of the Internship

The internship consisted of three technical tasks. They were carried out as separate pieces of work, using separate tools, and they are reported separately in this document. No

output of Task 1 or Task 2 feeds into Task 3, and no attempt has been made here to present them as stages of one pipeline, because they are not.

Task 1: OptiSystem SNS-TF-QKD Prototype

A physical-layer proof-of-concept implementation of an SNS-TF-QKD link in OptiSystem, with the protocol-level decision logic (window selection, send/not-send choice, decoy-intensity selection, phase randomisation) implemented through embedded MATLAB co-simulation blocks. The model uses fibre spans (SMF-28) between three stations, Alice, Bob and Charlie, at a laboratory-representative bit rate, and does not model a satellite channel or orbital geometry of any kind. Chapter 2 covers this task.

Task 2: Python Preprocessing Pipeline

A pipeline that reads the CSV files exported by the Task 1 OptiSystem model, aligns them by bit index, cleans up export artefacts, and computes the sifting, gain and error-rate quantities needed before a secure-key-rate estimate can be attempted. This pipeline consumes only OptiSystem-generated data. Chapter 3 covers this task.

Task 3: QSAFire

An independent Python simulation framework, QSAFire (Quantum Satellite Analysis Framework for Integrated Research and Evaluation), built to evaluate SNS-TF-QKD performance over a satellite-to-ground downlink. QSAFire models orbital geometry, ground-station visibility, atmospheric and turbulence loss, and a fully asymmetric SNS-TF-QKD secure-key-rate calculation, applied primarily to the Delhi-Mumbai ground-station pair and, secondarily, to three other Indian city pairs. QSAFire shares no code and no data path with Task 1 or Task 2. Chapter 4 covers this task in depth, and Chapter 5 carries its results forward into a comparison against the real Micius satellite-QKD mission.

Figure 1.1 summarises this separation. The three boxes are drawn with no connecting arrows between them for a reason: there is no data flow between them, and this report is organised to keep that boundary visible rather than to paper over it.

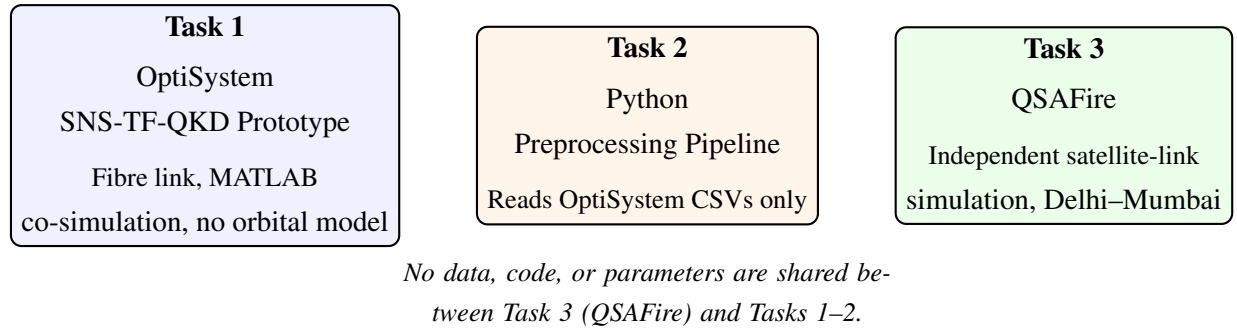


Figure 1.1: The three internship tasks and their independence. Task 1 and Task 2 form one path (OptiSystem output consumed by the Python preprocessing pipeline); Task 3 (QSAFire) is a separate, self-contained simulation study.

1.3 Objectives

The objectives set for the internship, task by task, were:

- Build and simulate a working SNS-TF-QKD link in OptiSystem at the physical layer, including MATLAB-driven protocol logic, and produce synchronised CSV logs from all three stations (Task 1).
- Write a Python pipeline capable of ingesting those CSV logs, correcting for common export issues, and computing the classical pre-processing quantities (sifted key length, gain, raw QBER) needed by a downstream key-rate calculation (Task 2).
- Independently evaluate, in software, whether an SNS-TF-QKD downlink between a low-Earth-orbit satellite and pairs of Indian ground stations is plausible under realistic atmospheric and turbulence conditions, and benchmark the resulting model against the published performance of the Micius satellite, the only satellite-QKD mission with public, peer-reviewed in-orbit data (Task 3, QSAFire).

1.4 Organisation of the Report

Chapter 2 documents the OptiSystem prototype: station-level architecture, MATLAB component roles, and simulation parameters. Chapter 3 documents the Python pre-processing pipeline built around that prototype’s CSV output. Chapter 4 documents QSAFire: its orbital, atmospheric, and turbulence models, the ground sites and links it was applied to, and a specific correction made to its turbulence model partway through the internship, together with the verified effect of that correction on the final results. Chapter 5 draws together QSAFire’s results across all four ground-station links and benchmarks the Delhi-Mumbai link against Micius. Chapter 6 closes the report with

an evidence-based summary of what was completed, its limitations, and a short account of reasonable next steps for each task.

1.5 Chapter Summary

This internship comprised three independent tasks under a shared broad theme, SNS-TF-QKD, but different scopes, tools, and levels of physical versus simulated implementation: an OptiSystem physical-layer prototype (Task 1), a Python CSV preprocessing pipeline for that prototype (Task 2), and QSAFire, an independent satellite-link simulation framework (Task 3). The remainder of this report treats each on its own terms and does not present them as one integrated system.

Chapter 2

Task 1: OptiSystem SNS-TF-QKD Prototype

2.1 Overview

This chapter describes the first of the three internship tasks: a physical-layer implementation of an SNS-TF-QKD link built in OptiSystem, with the protocol logic (window selection, the send/not-send decision, decoy-intensity selection, and phase randomisation) realised through MATLAB co-simulation blocks embedded in the schematic. The link involves three stations, Alice and Bob as the two transmitters and Charlie as the untrusted central measurement station, connected by single-mode fibre. The scope here is deliberately narrow: this is a laboratory-scale, fibre-based prototype used to verify that the SNS protocol’s logical structure can be built and made to run end to end inside OptiSystem, together with a consistent CSV logging scheme. It contains no orbital geometry, no satellite channel model, and no atmospheric turbulence model of the kind used in QSAFire (Chapter 4); any resemblance in vocabulary (fibre loss, detector efficiency) is coincidental to the fact that both tasks deal with a lossy optical channel, not evidence of a shared implementation.

2.2 Station Architecture

Figure 2.1 shows the top-level signal flow. Both Alice and Bob instantiate two parallel branches, labelled Z-window (the signal, key-generating basis) and X-window (the decoy, parameter-estimating basis), matching the SNS protocol’s basis structure directly. Each branch modulates its own continuous-wave laser: the Z-branch applies the SNS send/not-send decision, and the X-branch applies decoy-intensity selection followed by phase randomisation. Both branches recombine before entering a dedicated fibre span toward Charlie. At Charlie’s station, Alice’s and Bob’s fields interfere at a 50:50 coupler, one instance for each basis, and the two output ports of each coupler are monitored by a pair of single-photon detectors.

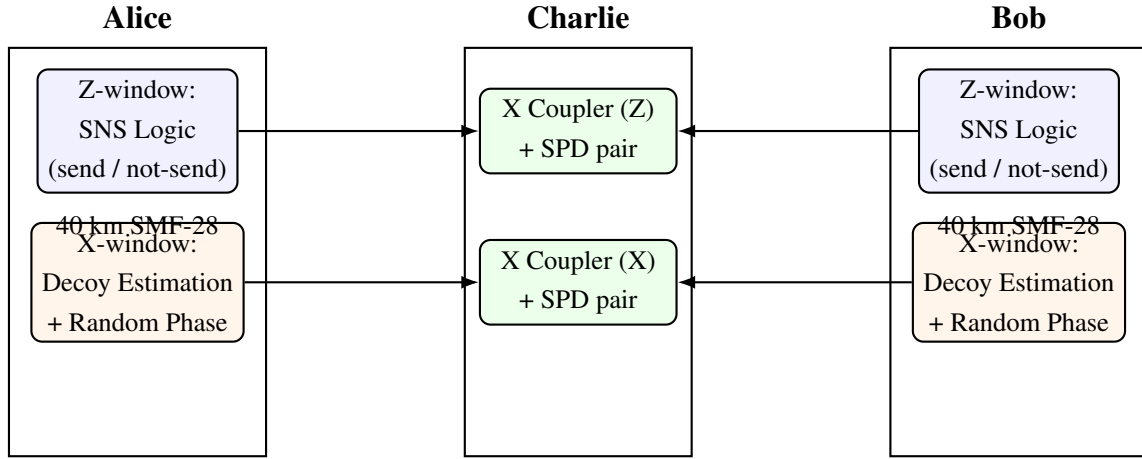


Figure 2.1: Top-level architecture of the Task 1 OptiSystem prototype. Alice and Bob each run independent Z-window and X-window branches; Charlie hosts one interference stage per basis, each followed by a pair of single-photon detectors. No orbital or satellite element is present anywhere in this diagram.

Bob's chain mirrors Alice's block for block, with one protocol-mandated difference that is easy to miss when copying one station's logic to build the other: Bob's Z-basis bit-value convention is the logical complement of Alice's. Alice logs bit value 1 for send and 0 for not-send; Bob logs bit value 0 for send and 1 for not-send [3]. This flip is what makes a valid Z-basis event show up, after Charlie's heralding click is correlated against both logs, as a bit *disagreement* between Alice and Bob rather than an agreement. Implementing Bob's SNS Logic block as a direct copy of Alice's, without flipping this convention, silently breaks the downstream error-rate calculation, since it corrupts the sign convention that any offline sifting stage would expect.

2.3 Electrical Domain and Window Selection

Each station's electrical front end generates two bit sequences: a pseudo-random bit sequence (PRBS) used purely to classify each time window as Z or X, and a second, independently configurable bit sequence used as a verification-friendly payload during debugging. Both are shaped into rectangular non-return-to-zero (NRZ) electrical waveforms at the system bit rate before driving any optical modulator. Window classification itself is done by a dedicated MATLAB block: for time window i , a PRBS bit of 1 routes that window down the Z-branch and a PRBS bit of 0 routes it down the X-branch. This block, and every other MATLAB block in the chain, must write exactly one log row per logical time window, not one row per oversampled waveform sample; getting this wrong (writing 32 rows per window instead of one, given the oversampling factor used here) is flagged in the source design documentation as the single most common implementation mistake in this class of architecture, and it was kept in mind throughout the build.

2.4 Protocol Logic via MATLAB Co-simulation

OptiSystem has no native block for a probabilistic, per-window branching decision, so every SNS-specific decision is implemented as a MATLAB co-simulation component acting as a programmable tap in the signal-flow graph. Table 2.1 summarises the role of each block used in this prototype.

Table 2.1: MATLAB co-simulation blocks implementing SNS protocol logic at each transmitting station.

Block	Role
Window Selection	Classifies each time window as Z (signal) or X (decoy) from the PRBS bit and routes the window to the corresponding branch.
SNS Logic (Z-branch)	Draws a Bernoulli(ϵ) decision per Z-window; commands the intensity modulator to pass a pulse at intensity μ' on “send” or to fully extinguish it on “not send”; logs the resulting Z-basis bit value under each station’s own convention.
Decoy Estimation (X-branch)	Selects one of the configured intensity classes (μ_2, μ_1 , vacuum) per X-window from a pre-agreed probability distribution, for later decoy-state parameter estimation.
Random Phase (X-branch)	Draws a uniform global phase $\delta \in [0, 2\pi)$ per window and drives the phase modulator; optionally quantises the draw into discrete phase slices for logging.
CSV logging (terminal)	Consolidates the per-window records from the blocks above, keyed by a common bit index, into the station’s final CSV file.

The sending probability ϵ and signal intensity μ' are the two free parameters that jointly determine the achievable key rate at a given loss; in this prototype they are exposed as configurable values rather than hard-coded, consistent with running them as a sweep in later iterations, although the runs reported here used a single fixed configuration rather than a distance sweep. The decoy-intensity selection block does not need to keep its draw private, since decoy-window phase and intensity information is publicly disclosed later in the protocol; the Z-branch send/not-send draw, by contrast, must stay private until sifting, and this is the one place in the whole chain where a logging bug would have actual security consequences rather than just a bookkeeping one [3].

2.5 Optical Domain and Charlie's Interference Stage

Every laser source in the model is a continuous-wave source at 193.1 THz (approximately 1552.52 nm), the ITU-T C-band reference channel, at 0 dBm output power. This wavelength sits inside the low-attenuation window of SMF-28 fibre, and the 0 dBm figure is a pre-attenuation reference power well above the single-photon regime; the actual reduction to signal, decoy, and vacuum intensity levels happens downstream, at the intensity modulators driven by the SNS Logic and Decoy Estimation blocks. A Mach-Zehnder modulator imprints the NRZ bit pattern as an amplitude on/off pattern ahead of this stage, and a phase modulator on the X-branch imprints the random phase drawn by the Random Phase block.

Both stations connect to Charlie over 40 km spans of SMF-28 fibre, one pair of spans for each of Alice and Bob (Z-branch and X-branch kept on physically separate fibre runs in this build, rather than multiplexed onto one). At 0.2 dB/km, a typical figure for SMF-28 near 1550 nm, each 40 km span contributes roughly 9 dB of loss, corresponding to a fibre transmittance of $\eta_{\text{fibre}} = 10^{-(0.2)(40)/10} \approx 0.126$. At Charlie, Alice's and Bob's fields are combined at a 50:50 coupler, one instance per basis, and the resulting output-port intensities depend on the relative phase between the two inputs; which of the two detectors registers a click is what carries information about the correlation between Alice's and Bob's independent decisions, exactly as required by the twin-field measurement principle [2]. Because the bit is inferred only from which detector clicks, and no logic downstream of the detectors is ever trusted with calibration information, the detectors themselves are treated purely as click/no-click loggers, consistent with the measurement-device-independent property this protocol family inherits [4].

Detection is modelled probabilistically rather than deterministically: a given window registers a click according to a detection efficiency η_{det} and a dark-count probability, both exposed as configurable detector parameters rather than fixed by the block itself. The literature on this protocol family uses a range of values for η_{det} depending on the detector technology being represented, from around 30% for a realistic InGaAs avalanche photodiode assumption up to 80% in some SNS numerical studies; this prototype leaves the exact operating point configurable rather than committing to one, since no specific detector hardware was being matched in this build.

2.6 Data Logging and Synchronisation

Every station accumulates its own CSV file over the course of a run: Alice.csv, Bob.csv, and a detector-side file at Charlie (split, in this build, into a Z-window and an X-window file). Every row in every file is keyed by a common bit index running from 0 to $N - 1$, because every MATLAB block, at both transmitters and at Charlie, is driven off the same

OptiSystem sample clock. This bit-index key is what makes a straightforward join between Alice’s, Bob’s, and Charlie’s logs valid downstream, and it is also what sidesteps the fibre’s real propagation delay (on the order of $196\text{ }\mu\text{s}$ one-way at 40 km): OptiSystem’s fibre block already applies the correct group delay to the waveform itself, so by the time Charlie’s blocks observe window i , they are observing the correctly delayed instance of that window, and no separate timestamp correction is needed as long as every block keys its log off the logical window counter rather than off raw simulation time.

A window that produces no click at either detector still receives a logged row with both click flags set to zero. This matters because, at this fibre loss and detector efficiency, the overwhelming majority of windows produce no click at all, and omitting no-click rows would corrupt any later attempt to compute a gain value, since that ratio’s denominator depends on the true total window count, not just the count of windows that happened to register a click.

2.7 Simulation Parameters

Table 2.2 lists the global parameters used for the runs reported in this chapter.

Table 2.2: Global simulation parameters for the Task 1 OptiSystem prototype.

Parameter	Value	Notes
Bit rate	10 Gbit/s	Sets the 100 ps time-window duration used for every per-window protocol decision
Sequence length	1024 bits	Per-run sequence length; a simulation-scale choice, not a protocol-scale one
Samples per bit	32	Oversampling factor for waveform representation
Sample rate	320 GSa/s	Bit rate $\times$ samples per bit
Time window (per run)	102.4 ns	Sequence length / bit rate
Laser frequency	193.1 THz ($\approx 1552.52\text{ nm}$)	ITU-T C-band reference channel
Laser output power	0 dBm	Pre-attenuation reference power
Fibre type and length	SMF-28, 40 km per span (4 spans total)	Laboratory-scale placeholder, not a long-haul distance
Assumed fibre attenuation	0.2 dB/km	Standard value for SMF-28 near 1550 nm; yields $\approx 9\text{ dB}$ loss per span

A sequence length of 1024 bits, at 32 samples per bit, gives exactly 32768 waveform samples, a convenient power of two for OptiSystem’s internal frequency-domain operations and for indexing inside the MATLAB blocks. It is, however, far too short to obtain a statistically meaningful decoy-state bound or a finite-key secure-key-rate estimate on its own; a single run at this length verifies that the architecture behaves correctly window by window, not that the resulting key rate is representative of the protocol’s real performance at scale. This point is picked up again in Section 2.8.

2.8 Limitations of the Prototype

Three limitations are worth stating plainly rather than leaving implicit. First, the 40 km fibre spans used here are a tractable, laboratory-representative distance, not an attempt to demonstrate SNS-TF-QKD’s headline long-distance advantage, which only becomes apparent over spans of several hundred kilometres; extending this build to that regime would also require an explicit active phase-drift compensation stage at Charlie, since phase drift accumulates with distance and this build does not model that compensation loop as a separate component. Second, the 1024-bit sequence length is adequate for verifying that the protocol’s logical structure runs correctly end to end, but not for producing a statistically defensible secure-key-rate figure; doing that would require either a much longer sequence or repeated runs whose CSV logs are concatenated before any offline analysis is attempted, which is the exact hand-off point to Task 2 (Chapter 3). Third, this prototype does not include any satellite, orbital, or atmospheric turbulence modelling of any kind; any surface-level resemblance to the channel-loss language used in Chapter 4 is a consequence of both tasks dealing with a lossy optical channel, not a sign that the two share code or data.

2.9 Chapter Summary

Task 1 built a working, three-station SNS-TF-QKD prototype in OptiSystem, with the SNS protocol’s window-selection, send/not-send, decoy-intensity, and phase-randomisation logic implemented as MATLAB co-simulation blocks, and with every station logging a bit-index-synchronised CSV file suitable for offline analysis. The build is explicitly a fibre-based, laboratory-scale proof of concept: it establishes that the protocol’s logic runs correctly inside OptiSystem and produces consistent, well-synchronised logs, and it stops there. No secure-key-rate figure is claimed from this task alone, and no satellite or orbital element is present in it; the CSV output it produces is the input to Task 2, described next.

Chapter 3

Task 2: Python Preprocessing Pipeline

3.1 Overview and Scope

This chapter documents the second internship task: a Python pipeline that reads the CSV files exported by an OptiSystem-style SNS-TF-QKD simulation run, synchronises them, and carries out the classical post-processing that Wang et al.’s SNS security analysis requires, namely basis separation, phase post-selection, decoy-state parameter estimation, Z-basis gain and QBER, and the secret-key-rate calculation itself [3]. The pipeline consumes five CSV files:

```
alice_signal_record.csv
bob_signal_record.csv
charlie_record_x.csv
charlie_record_z.csv
charlie_xbasis_log.csv
```

It is, deliberately, only a consumer of this data. It shares no code and no execution path with QSAFire (Chapter 4); the two happen to both compute a secure key rate, but from entirely different inputs and for entirely different channel models.

The version reported here is a single, linearly-readable notebook rather than a modular package. Every quantity computed is tied to a specific numbered equation in Wang et al., and no curve-fitting or empirical adjustment is used anywhere; where a rate is estimated from a finite count, a 95% Wilson score confidence interval is reported alongside it rather than a bare point estimate.

3.2 Input Data and Synchronisation

The CSV set exercised in this task contains 32,768 synchronised time windows, a considerably longer run than the single 1024-window example parametrised in Chapter 2, and consistent with that chapter’s own observation that a short run is adequate for verifying protocol logic but not for producing a statistically meaningful decoy-state bound. Table 3.1 summarises the synchronisation check performed before any analysis: every file has the same row count, no duplicate window indices, and no missing required columns; the only null cells present are in Alice’s and Bob’s records, in columns that are legit-

imately blank for a given window (for instance, decoy-related fields on windows that turned out to be Z-windows).

Table 3.1: Synchronisation summary across the five input CSV files.

File	Rows	Duplicate index	Missing columns
alice_signal_record.csv	32,768	0	0
bob_signal_record.csv	32,768	0	0
charlie_record_x.csv	32,768	0	0
charlie_record_z.csv	32,768	0	0

All four files share the same window index range, so the common and union window sets are identical (32,768 windows each), and the merge on `Window_Index` used throughout the rest of the pipeline is a clean inner join with no dropped rows. Alice’s and Bob’s independently-chosen basis labels (Z or X) agree on 52.93% of windows (17,344 of 32,768). This is expected, not a fault: Alice and Bob choose their signal/decoy window for a given index independently, so roughly half of all windows are expected to mismatch, and the SNS protocol itself discards these as “mismatching windows” rather than attempting to use them [3].

3.3 Basis Separation

Only basis-matched windows are retained for further analysis. Table 3.2 shows the resulting split.

Table 3.2: Window counts after discarding basis-mismatched windows.

Category	Count	Fraction of total
Total windows	32,768	100.00%
X windows (basis-matched)	9,536	29.10%
Z windows (basis-matched)	7,808	23.83%
Mismatched (discarded)	15,424	47.07%

3.4 X-Basis Phase Post-Selection

Within X-windows, the SNS protocol requires a further match: Alice and Bob must have used the same decoy intensity for that window to count as a true X_μ window, and, for

non-vacuum intensities, their phases must satisfy the acceptance condition

$$1 - |\cos(\delta_A - \delta_B)| \leq |\lambda| \quad (3.1)$$

which is Eq. (1) of Wang et al. The simulated data discretises phase into M slices of width $\Delta = 2\pi/M$; the pipeline verifies this slice-assignment formula directly against the raw phase column before using it (confirmed exact on 100% of Alice’s X-window rows, with $M = 16$, $\Delta = 0.3927$ rad, and an effective $\lambda_{\text{eff}} = 1 - \cos \Delta = 0.0761$), and uses slice equality as the operational stand-in for Eq. (3.1). The discrete rule and the continuous Eq. (3.1) criterion agree on 81.30% of non-vacuum windows, which is the expected discretisation edge effect (two phases can share a slice while sitting on opposite sides of the λ_{eff} boundary, and vice versa) rather than an implementation fault.

Of the 9,536 basis-matched X-windows, 3,451 (36.19%) have matching decoy intensity between Alice and Bob (1,570 at the higher decoy intensity, 1,511 at the lower, 370 at vacuum). Restricting to the 3,081 non-vacuum, intensity-matched windows and applying the phase-acceptance rule leaves 180 accepted windows, giving an acceptance probability $P_{\text{acc}} = 0.0584$ (95% Wilson CI: [0.0507, 0.0673]), close to the $1/M = 0.0625$ expected for uniformly random, independently drawn phases. This closeness is itself a useful sanity check: it indicates the phase draws in this dataset behave as the protocol assumes, even though, as Section 3.9 discusses, the detector clicks associated with those phases do not.

3.5 X-Basis Detection Analysis

An effective X-basis event requires exactly one of Charlie’s two detectors to click; double clicks and no-clicks are not effective events. Of the 180 phase-matched windows, 34 produced a single-detector click (15 on detector 1, 19 on detector 2), 2 produced a double click, and 144 produced no click at all, giving an X-basis gain $Q_X = N_{\text{click}}/N_{\text{accepted}} = 0.1889$ (95% Wilson CI: [0.1384, 0.2523]).

The X-basis detector error rate follows Wang et al.’s sign convention on $\cos(\delta_B - \delta_A)$: a detector-1 click with a negative cosine, or a detector-2 click with a non-negative cosine, counts as a wrong bit. On this dataset, 19 of the 34 clicks are wrong by that rule, giving $E_X = 0.5588$ (95% Wilson CI: [0.3945, 0.7112]). This is a small sample by any standard, 34 clicks in total, and the width of the confidence interval reflects that directly; it is flagged here rather than treated as a precise number.

3.6 Decoy-State Yield and Phase-Error Estimation

The decoy-state analysis (Eqs. 44 and 45 of Wang et al.) uses the click rate at each intensity level across *all* intensity-matched X-windows, not just the phase-matched subset, since a phase-randomised coherent state’s click rate does not depend on the announced phase. Table 3.3 gives the per-intensity statistics.

Table 3.3: Per-intensity X-window click statistics used in the decoy-state estimate.

Intensity level	μ	Windows	Click rate
Vacuum (μ_0)	0.00	370	0.1514
Decoy 1 (μ_1)	0.05	1,511	0.1760
Decoy 2 (μ_2)	0.20	1,570	0.1561

Applying Eq. (44) to these three points gives a single-photon yield lower bound $s_1 = 0.5082$, and applying Eq. (45), using the phase-matched click-error rate at μ_1 ($E_{\mu_1}^X = 10/18 = 0.5556$), gives a phase-error upper bound $e_1^{ph} = 0.6378$. Both values are reported exactly as computed; e_1^{ph} in particular is far higher than a workable SNS-TF-QKD link would tolerate, and Section 3.9 discusses why.

3.7 Z-Basis SNS Categories, Gain and QBER

Every Z-window falls into one of four categories depending on Alice’s and Bob’s independent send/not-send decisions. Table 3.4 shows the observed split.

Table 3.4: Distribution of Z-window SNS categories.

SNS category	Count	Fraction of Z windows
Alice sends, Bob does not	1,574	20.16%
Bob sends, Alice does not	1,642	21.03%
Both send	682	8.73%
Both do not send	3,910	50.08%

Only the first two categories carry genuine key bits; the “both send” and “both not-send” categories produce a bit error whenever an effective event occurs on them, since Alice’s and Bob’s bit-value conventions necessarily disagree in that case. Of the 7,808 Z-windows, 1,203 produced an effective click, giving a Z-basis gain $Q_Z = 0.1541$ (95% Wilson CI: [0.1462, 0.1623]); of those clicks, 699 fall in the both-send or both-not-send categories, giving QBER = 0.5810 (95% Wilson CI: [0.5530, 0.6086]).

3.8 Secret Key Rate

The per-window secret key rate is computed from Wang et al.'s Eq. (4):

$$R = 2\varepsilon(1 - \varepsilon) \mu' e^{-\mu'} s_1 (1 - H(e_1^{ph})) - S_Z f H(E^Z) \quad (3.2)$$

where ε is the average observed sending probability, μ' the mean signal intensity actually used on send decisions, $S_Z = Q_Z$, $E^Z = \text{QBER}$, $f = 1.1$ the error-correction inefficiency factor, and $H(\cdot)$ the binary entropy function. Table 3.5 lists every constituent term as computed by the pipeline.

Table 3.5: Constituent terms of the Eq. (4) secret-key-rate calculation.

Quantity	Value
ε (Alice 0.2889, Bob 0.2976, averaged)	0.2933
μ' (mean signal intensity on send)	5.899×10^{-3}
s_1 (single-photon yield, Eq. 44)	0.5082
e_1^{ph} (phase-error bound, Eq. 45)	0.6378
$H(e_1^{ph})$	0.9445
$S_Z = Q_Z$	0.1541
$\text{QBER} = E^Z$	0.5810
$H(E^Z)$	0.9810
f	1.1
R (Eq. 4, per window)	-0.1662

Figure 3.1 plots the four basis-level rates together. All four sit close to or above 0.5, which is the immediate visual signal that something in the channel statistics, not the analysis, is off; a workable SNS-TF-QKD link needs QBER and e_1^{ph} well below this level, not near it.

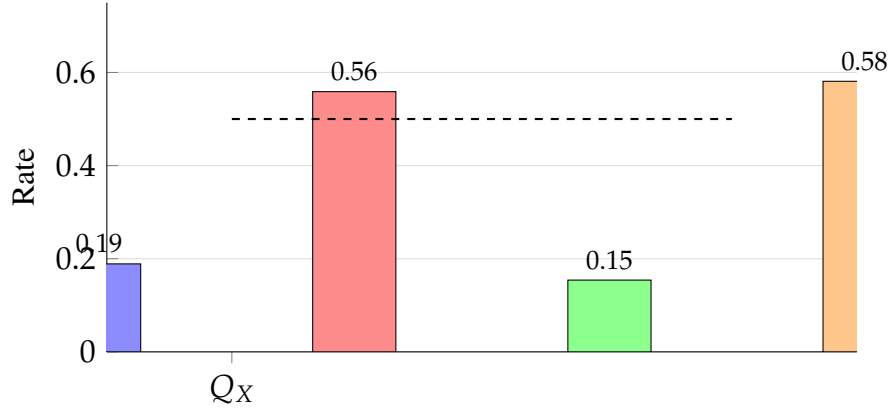


Figure 3.1: Basis-level gain and error rates computed by the pipeline. The dashed line marks 0.5; both error rates (E_X and QBER) sit at or above it, well outside a range in which a positive key rate would be expected.

As a cross-check, the pipeline also evaluates the algebraically equivalent bit-count form, Eq. (3), reconstructing n_1 from $s_1 \cdot \mu' e^{-\mu'} \cdot N_{Z1}$ and n_t from the observed Z-basis click count. This gives $N_f = -1297.6$ bits over the run, or $N_f/N_{\text{total}} = -0.0396$ per window, in the same direction (negative) as Eq. (4) but not equal in magnitude to $R = -0.1662$. Both are reported as computed rather than reconciled to force agreement; the discrepancy is most plausibly a consequence of approximating n_1 from an aggregate yield and Poisson factor rather than from a directly measured single-photon count, and is noted here as an open item for the next iteration of the pipeline rather than papered over.

3.9 Honest Assessment of the Results

The pipeline runs end to end without error and produces a negative secret key rate on this particular CSV set. Two observations point to the input data, not the analysis code, as the reason. First, the phase-matched X-basis sample is small, 34 clicks overall and 18 at the lower decoy intensity specifically, once basis-matching, intensity-matching, and the roughly 1-in-16 phase-acceptance probability are all applied in sequence; the resulting Wilson intervals are wide enough that E_X and e_1^{ph} carry substantial statistical uncertainty on top of their already-high point estimates. Second, and more fundamentally, the detector clicks in the Charlie-side CSVs do not appear to be statistically coupled to the phase difference $\cos(\delta_B - \delta_A)$ or to the SNS send/not-send category in the way a physically simulated interference channel would produce; QBER and E_X landing close to 0.5, the value expected from an uncorrelated coin flip, is consistent with this. Taken together, this looks like a CSV set that is structurally correct (every column, every join, every count behaves exactly as the protocol specification predicts) but not calibrated to represent a physically sensible channel.

This distinction matters for how the result should be read. It is a statement about this

specific dataset, not a limitation of the pipeline: the same code, run against a CSV set in which detector clicks are properly coupled to the underlying interference physics, is expected to produce a small, positive e_1^{ph} and a positive key rate, consistent with published SNS-TF-QKD numerical results [3]. Validating the pipeline against such a calibrated run is the natural next step before drawing any conclusion from these numbers beyond exercising the code correctly.

3.10 Chapter Summary

Task 2 built a Python pipeline that ingests OptiSystem-style CSV exports, synchronises them by window index, separates and phase-post-selects the X-basis and Z-basis populations, runs the Wang et al. decoy-state and phase-error estimators, and computes the Eq. (4) secret key rate, with a Wilson confidence interval reported on every rate estimated from a finite count. On the CSV set exercised here, the pipeline returns a negative key rate, traced to detector clicks that do not appear to be statistically coupled to the underlying phase and send/not-send variables, a property of this particular dataset rather than a defect in the pipeline itself. The pipeline is not connected to QSAFire (Chapter 4) in any way; it operates purely on OptiSystem-style classical logs and stops at the secure-key-rate figure for a single simulated run.

Chapter 4

Task 3: QSAFire, the Satellite-Link Simulation Framework

4.1 Overview

QSAFire (Quantum Satellite Analysis Framework for Integrated Research and Evaluation) is an independent Python framework built to answer one question: is an SNS-TF-QKD downlink between a low-Earth-orbit satellite and a pair of Indian ground stations plausible once realistic orbital geometry, atmospheric loss, and turbulence are accounted for, and how does the answer compare with what has actually been demonstrated in orbit. It shares no code, no data, and no CSV format with Task 1 or Task 2; where this chapter and Chapter 3 both mention quantities like channel transmittance or a secure key rate, they refer to two unrelated calculations over two unrelated channel models, one a 40 km fibre link, the other a satellite-to-ground free-space link several hundred to over a thousand kilometres long.

QSAFire was applied to six Indian ground sites, Delhi, Mumbai, Nainital, Ahmedabad, Bangalore, and Hyderabad, and to four candidate links between pairs of them. Delhi-Mumbai is the primary case study and the only link carried through to a full seasonal, weather-gated annual model; the other three links (Delhi-Nainital, Mumbai-Ahmedabad, Bangalore-Hyderabad) are evaluated to the same geometry, atmosphere, turbulence, and key-rate depth at a single representative best-elevation pass. This chapter describes the models QSAFire is built from; the results across all four links, and the comparison against the real Micius mission, are given in Chapter 5.

4.2 Orbital Geometry Model

The satellite is placed in a 500 km circular, sun-synchronous orbit. Rather than assuming a sun-synchronous inclination, QSAFire solves for it from the J_2 secular nodal-precession condition, i.e. the inclination is a computed output of the orbital mechanics, not a value copied from a reference mission. Before being trusted at the Micius-like 500 km altitude, the solver is checked against two other real, operating sun-synchronous missions: it reproduces Landsat-8/9's inclination (98.2°) and Sentinel-2's inclination (98.6°) to within

0.1° at their respective altitudes, which is treated as an independent correctness check on the solver before it is used to predict anything about the Delhi-Mumbai geometry itself. At 500 km, the solver returns an inclination of 97.40° and a nodal period of 94.6 to 94.7 minutes, around 15.2 orbits per day.

The elevation-to-slant-range relationship that this geometry feeds into every downstream loss calculation is validated directly against two real, published Micius data points, rather than only self-consistently against the orbital model. Table 4.1 shows both checks.

Table 4.1: Orbital and elevation-range validation against real reference missions.

Quantity	QSAFire	Reference	Deviation
Inclination at Landsat-8/9 altitude	solver output	98.2° (real)	$<0.1^\circ$
Inclination at Sentinel-2 altitude	solver output	98.6° (real)	$<0.1^\circ$
Slant range @ 85.7° elevation	501.31 km	507.0 km [5]	-1.12%
Slant range @ 25.0° elevation	1,031.94 km	1,034.7 km [5]	-0.27%

Both slant-range checks pass a hard assertion built into the code requiring agreement to within 2.0%; the actual deviations are well under 1.2%. This is the most directly literature-anchored validation in the whole framework, since orbital mechanics at this level is a solved, no-free-parameters problem: if the geometry did not reproduce Micius’s own reported numbers, nothing computed downstream of it would be trustworthy regardless of how the atmosphere or turbulence were modelled.

4.3 Ground Sites

Table 4.2 lists the six ground sites QSAFire was applied to, with the aerosol optical depth (AOD) and turbulence multiplier each site is carried with.

Table 4.2: Ground site characteristics used by QSAFire.

Site	Alt. (m)	AOD ₅₀₀	Ångström α	Turb. mult.	Confidence
Delhi (Alice, primary)	216	0.78	0.78	1.00	High [6]
Mumbai (Bob, primary)	14	0.45	0.90	1.00	High [7]
Nainital (ARIES)	1,951	0.18	1.00	0.40	Medium (proxy)
Ahmedabad	53	0.40	1.00	1.00	High [8]
Bangalore	920	0.25	1.00	1.00	High [9, 10]
Hyderabad	542	0.55	0.90	1.00	Medium [11]

Delhi carries the heaviest aerosol burden of the six sites by a wide margin, driven by its well-documented winter haze problem, rising to AOD 0.91 in winter conditions specifically [6, 12]. Nainital is the outlier at the other end: it is the only site with a turbulence multiplier below 1, and this is carried at Medium confidence, not High, because the supporting citation [13] is a radar-derived turbulence estimate, a qualitative proxy for the optical C_n^2 that the Rytov-variance calculation actually needs, not a direct optical measurement. QSAFire’s documentation treats Nainital’s real-world status as a working astronomical observatory site as independent, non-literature corroboration of this multiplier: large telescopes are not sited at locations with poor seeing, so the site-selection fact itself is weak evidence in the same direction as the radar proxy. No site in this table has a direct optical ground-level C_n^2 measurement backing its turbulence multiplier; this is flagged again in Section 4.10 as the single largest open item in the whole framework.

4.4 Atmospheric Loss Model

Atmospheric transmittance is computed from a Rayleigh-scattering baseline plus each site’s AOD, integrated along the actual slant path rather than assumed constant with elevation. The Rayleigh component alone is cross-checked against a standard reference value: QSAFire computes a Rayleigh zenith optical depth of 0.1434 at 500 nm against the AMS Glossary’s reference figure of 0.145, a -1.09% deviation. Table 4.3 gives the resulting zenith and pass-window loss for the two primary-link stations.

Table 4.3: Atmospheric loss at 810 nm, Delhi and Mumbai.

Site	τ_{zenith}	Transmitt.	Zenith loss (dB)	Pass loss (dB)
Delhi	0.576	56.2%	2.50	3.03 – 7.25
Mumbai	0.267	76.6%	1.16	1.42 – 3.36
Delhi, winter-haze	0.689	50.2%	2.99	(sensitivity case)

Delhi’s atmosphere costs roughly twice the dB loss of Mumbai’s at every point across the pass, consistent with its higher AOD, and this gap persists through every downstream link that includes Delhi as an endpoint. Both stations’ combined atmosphere-plus-turbulence loss at their best-elevation point (3.03 dB Delhi, 1.42 dB Mumbai) sits below Micius’s own reported 3 to 8 dB range at 1,200 km [5], which is expected given that Delhi-Mumbai’s best-case slant range, around 600 to 800 km, is considerably shorter than Micius’s reported 1,200 km figure.

4.5 Turbulence Model and the Uplink Correction

4.5.1 Initial model and the identified gap

Optical turbulence is quantified through the Rytov variance, which sets the scintillation-index regime (weak, moderate, or strong fading) that the aperture-averaged Gamma-Gamma fading model downstream is parametrised by. QSAFire’s original slant-path implementation used the standard plane-wave Rytov-variance integral,

$$\sigma_R^2 = 2.25 k^{7/6} \sec^{11/6}(\zeta) \int C_n^2(h) (h - h_0)^{5/6} dh, \quad (4.1)$$

which correctly reduces, in the constant- C_n^2 horizontal-path limit, to the textbook plane-wave form $\sigma^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$, and this reduction check passes in the notebook. The gap identified partway through the internship was not in this formula itself, but in its applicability: SNS-TF-QKD over a satellite link is an uplink geometry (ground station transmitting up to the satellite), for which the physically appropriate quantity is the *spherical-wave* Rytov variance, not the plane-wave one used in Eq. (4.1). A satellite downlink and a satellite uplink see different effective scintillation for the same path, because a spherical wave undergoes a degree of self-healing across the propagation path that a plane wave does not.

4.5.2 Two unverified attempts, and why they were rejected

Two candidate fixes were tried and discarded before the correct one was confirmed, and both are recorded here because the reason they were wrong is instructive. The first attempt changed the coefficient and exponent in Eq. (4.1) to $1.23 \sec^{5/6}(\zeta)$ from an unverified recollection that spherical-wave slant formulas simply swap the exponent from $11/6$ to $5/6$; this was not checked against any source before being implemented, and the notebook’s own internal consistency check (the same reduction-to-horizontal-form test used to validate Eq. (4.1)) immediately flagged a 45% disagreement. The second attempt substituted a coefficient of 0.56 in its place, again from memory rather than a checked source, and was equally unreliable; a contradictory intermediate result produced in the course of that attempt went unnoticed at the time, which is itself a reminder that a plausible-looking derivation is not a substitute for checking against a citable formula. Both attempts are noted here rather than silently dropped, since the eventual fix is only trustworthy because it replaced this guess-and-check process with an actual literature lookup.

4.5.3 The verified fix

The standard relationship between plane-wave and spherical-wave Rytov variance for a uniform horizontal path, given in Andrews' *Field Guide to Atmospheric Optics* [14], is

$$\sigma_{\text{plane}}^2 = 1.23 C_n^2 k^{7/6} L^{11/6}, \quad \beta_0^2 = 0.5 C_n^2 k^{7/6} L^{11/6}, \quad (4.2)$$

giving a constant conversion ratio $0.5/1.23 = 0.4065$ between the two. This exact ratio is used as a slant-path conversion factor in the peer-reviewed satellite/balloon-QKD turbulence literature [15], and was adopted here as a citable, verifiable simplification, in preference to a full geometry-dependent weighting function inside the path integral, which the literature indicates exists but which could not be verified cleanly enough from available sources to use with confidence. The implemented fix leaves the plane-wave path integral, Eq. (4.1), unchanged, and multiplies its output by 0.4065 as a final step; a second validation stage was added confirming that the corrected output reduces to the textbook spherical-wave horizontal value (the 0.5 coefficient) in the same constant- C_n^2 , zenith limit used to validate the original formula. Both validation stages pass, with a numerical-integration artefact of -0.222% , the same order of magnitude as the original check.

4.5.4 Verified effect on the results

The corrected notebook was executed end to end (all 250 cells, zero errors) and compared against a full execution of the original, uncorrected version, not against numbers quoted from memory. Table 4.4 shows the turbulence regime before and after correction for the two cases where the change is largest.

Table 4.4: Effect of the spherical-wave correction on turbulence regime classification.

Site / case	Regime, original	Regime, corrected
Delhi, site case ($\times 2$ urban enhancement)	moderate 4%, weak 96%	weak 100%
Delhi, shared baseline	weak 100%	weak 100%
Mumbai, both cases	weak 100%	weak 100%

Delhi's site case is the one genuine regime reclassification produced by the fix: pre-correction, it crossed $\sigma_R^2 = 1$ near the low-elevation edge of the pass, placing 4% of the pass window in the moderate-fading regime; post-correction, the same case stays below $\sigma_R^2 \approx 0.44$ throughout, entirely inside the weak-fading regime. The Fried parameter computed under the shared HV 5/7 turbulence profile is validated separately against its canonical zenith value: 4.96 cm computed against a 5.00 cm target, a -0.80% deviation.

Despite this substantial change in the turbulence numbers themselves, the final secure-key-rate figures move very little, because turbulence fading was never the binding constraint on these links to begin with (Section 4.7 explains why). Table 4.5 gives the before/after comparison for all four links.

Table 4.5: Effect of the turbulence correction on final secure key rate and breakeven suppression, all four links.

Link	SKR orig. (kbit/s)	SKR corr. (kbit/s)	Change	Breakeven orig.	Breakeven corr.
Delhi–Mumbai	0.0940	0.0947	+0.74%	841.5×	854.2×
Delhi–Nainital	0.3298	0.3300	+0.06%	162.6×	162.8×
Mumbai–Ahmedabad	0.2908	0.2911	+0.10%	238.3×	238.9×
Bangalore–Hyderabad	0.2823	0.2831	+0.28%	244.4×	245.9×

In short: the underlying physics was wrong before the fix and is verified correct after it, the turbulence numbers themselves change substantially in places, but the headline conclusions about which link performs best and what suppression margin is needed do not change in any way that would have altered a decision made from the pre-correction figures. The one place the fix does change a substantive conclusion, not just a number, is Mumbai’s seasonal availability, discussed in Chapter 5.

4.6 Pointing Loss

Two pointing-loss cases are carried through the framework side by side rather than collapsed into one number. The first uses Micius’s own published in-flight pointing jitter, $0.47\text{ }\mu\text{rad}$, giving a computed pointing loss of -0.038 dB , comfortably inside Micius’s own stated pointing-loss budget of under 3 dB [5]. The second is a more conservative, design-oriented default of $2.5\text{ }\mu\text{rad}$ drawn from the recent satellite-QKD pointing-loss literature [16], giving -0.969 dB . The first case represents demonstrated, real flight performance; the second is a deliberately pessimistic planning case, and both are reported rather than choosing one and discarding the other.

4.7 Background Light and the Real Binding Constraint

A diagnostic run isolating each loss and noise term in turn, at each station’s best-elevation point, shows that geometry, atmosphere, turbulence, pointing, and detector efficiency are all individually adequate; what closes the Delhi–Mumbai link is night-sky background light. With dark counts as the only noise source, both stations return a positive per-pulse rate (Delhi $+1.80 \times 10^{-5}$ bits/pulse, Mumbai $+2.60 \times 10^{-5}$ bits/pulse); adding

realistic night-sky background alone flips both negative (Delhi -2.99×10^{-5} , Mumbai -3.63×10^{-5}). This is the single most important structural finding in the framework, and it is unchanged by the turbulence correction in Section 4.5: turbulence was never what was breaking these links.

The background-suppression factor required to break even, under the corrected turbulence model and with a realistic misalignment parameter $e_a = 0.15$, is $407\times$ for Delhi and $209\times$ for Mumbai. Realistic misalignment compounds the idealised, misalignment-free background-only requirement (11 to $22\times$) by roughly a further factor of 20 , because the two noise budgets interact rather than adding independently. Both final figures remain well inside real demonstrated capability: portable ground stations have already achieved background suppression on the order of $10^{12}\times$ (120 dB) using combined spatial and spectral filtering [17], over 100 times more suppression than either station in this study actually needs. QSAFire adopts a $3\times$ margin over the computed breakeven threshold as its design-safety convention for the results reported in Chapter 5; this margin is a convention carried over from earlier in the project, not re-derived from first principles here.

4.8 SNS-TF-QKD Rate Model for the Satellite Link

The satellite-link secure-key-rate calculation is a fully asymmetric generalisation of the SNS-TF-QKD rate formula introduced in Chapter 3 (Eq. (3.2)): Alice’s and Bob’s channel transmittances η_A, η_B and turbulence variances $\sigma_{R,A}^2, \sigma_{R,B}^2$ are kept distinct rather than assumed equal, since the two ground stations in a given link are never at identical elevation or atmospheric conditions during a real pass; each arm’s fading is modelled independently through the Gamma-Gamma distribution parametrised by that arm’s own Rytov variance; and the free protocol parameters, the sending probability ε and the two decoy intensities μ_1, μ_2 , are jointly optimised per link rather than fixed to one shared value across all four links. The breakeven background-suppression threshold introduced in Section 4.7 is likewise re-derived per link rather than assumed constant, since it depends on that link’s own loss budget. The resulting per-link master table, and the ranking it produces, are given in Chapter 5.

4.9 Software Architecture

Figure 4.1 summarises the module-level flow of the framework, from orbital geometry through to a per-link secure-key-rate figure. Every stage carries at least one validation check against an external, citable reference before its output is passed to the next stage; this is a deliberate design choice, since an error introduced early (as Section 4.5 shows) propagates through every later stage silently unless it is caught close to its source.

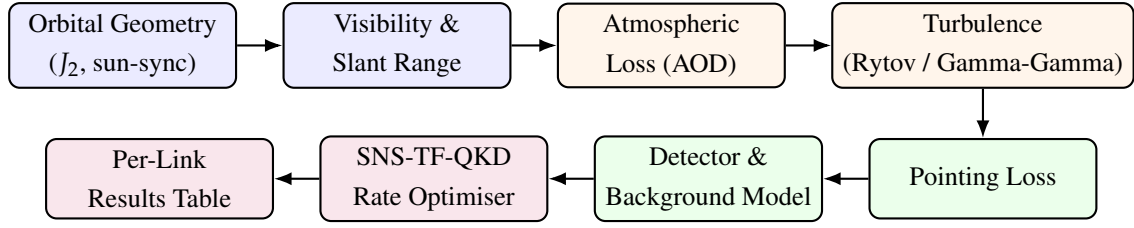


Figure 4.1: QSAFire’s module-level processing pipeline. Each stage is validated independently against a citable external reference (orbital solver against Landsat-8/9, Sentinel-2 and Micius; atmosphere against the AMS Glossary Rayleigh figure and Micius’s reported loss budget; turbulence against the HV 5/7 canonical Fried parameter) before its output feeds the next stage.

4.10 Limitations and Confidence Caveats

The following are carried forward unchanged by the turbulence correction and remain open at the end of this task:

- No real, optical, ground-level C_n^2 measurement exists for any of the six sites. Every turbulence multiplier beyond the shared HV 5/7 baseline (Delhi’s $\times 2$ urban case, Nainital’s $\times 0.4$) is a proxy or an indicative estimate, not a direct optical measurement, and this is the single largest open item in the framework.
- No real background-light measurement exists for any city. Delhi’s and Mumbai’s *relative* night brightness is grounded in a real study [18], but the *absolute* suppression multipliers used in Section 4.7 are indicative, not measured.
- Nainital’s, Ahmedabad’s, Bangalore’s, and Hyderabad’s atmospheric parameters are sourced from the literature at High or Medium confidence, but none were measured specifically for this application.
- The $3\times$ adopted suppression margin (Section 4.7) is a design convention, not independently derived in this work.
- Only Delhi-Mumbai carries a full seasonal, weather-gated annual model; the other three links are evaluated at a single representative best-elevation pass.
- The 0.4065 spherical-wave conversion factor (Section 4.5) is a constant-ratio approximation anchored to the horizontal-path limit, not a full geometry-dependent slant-path derivation; a fully rigorous treatment would use a geometry-dependent weighting function instead, and this remains a legitimate follow-up.
- The early, generic terrestrial free-space-optical illustration used elsewhere in the framework still uses the plane-wave Rytov formula; this was left as is because that

illustration is unrelated to the satellite architecture, not because it was re-verified to be already correct.

4.11 Chapter Summary

QSAFire is an independent framework covering orbital geometry, atmospheric loss, turbulence, pointing loss, and a fully asymmetric SNS-TF-QKD rate calculation, applied to six Indian ground sites and four candidate links, with the orbital solver and elevation-range formula validated directly against real Micius, Landsat, and Sentinel data. A specific error in the turbulence model, use of the plane-wave rather than spherical-wave Rytov variance for an uplink geometry, was identified, initially mis-corrected twice using unverified assumptions, and then fixed properly using a citable literature ratio, with the fix verified by executing the corrected notebook end to end rather than asserted from a partial read. The corrected turbulence numbers differ substantially from the originals in places, including one genuine regime reclassification, but the final key-rate and suppression-margin conclusions are essentially unchanged, because background light, not turbulence, is what actually limits these links. Chapter 5 presents the resulting per-link and per-site results and the comparison against Micius's own published performance.

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents QSAFire’s results across all four links, the resulting site and link rankings, the Delhi-Mumbai seasonal model, and the comparison against Micius’s own published in-orbit performance. All figures below are read directly from the corrected (spherical-wave/uplink) notebook described in Chapter 4, executed end to end with zero errors, and are not recomputed by hand for this chapter.

5.2 Per-Link Final Performance

Table 5.1 gives the geometric and optical parameters for each of the four links, and Table 5.2 gives the resulting security parameters and final secure key rate. Both tables come from the same fully asymmetric, jointly-optimised calculation described in Section 4.8.

Table 5.1: Link geometry and per-arm transmittance/turbulence.

Link	Dist. (km)	η_A	η_B	$\sigma_{R,A}^2$	$\sigma_{R,B}^2$
Delhi–Mumbai	1,149.4	7.77×10^{-4}	1.15×10^{-3}	0.134	0.135
Delhi–Nainital	234.5	2.45×10^{-3}	3.86×10^{-3}	0.058	0.053
Mumbai–Ahmedabad	440.5	2.74×10^{-3}	2.86×10^{-3}	0.065	0.066
Bangalore–Hyderabad	500.6	3.04×10^{-3}	2.41×10^{-3}	0.069	0.069

Table 5.2: Link security parameters and final secure key rate.

Link	Breakeven	Adopted ($3\times$)	QBER	Phase err.	SKR (kbit/s)
Delhi–Mumbai	$854.2\times$	$2,562.6\times$	2.46%	15.1%	0.0947
Delhi–Nainital	$162.8\times$	$488.4\times$	2.39%	15.1%	0.3300
Mumbai–Ahmedabad	$238.9\times$	$716.7\times$	2.39%	15.1%	0.2911
Bangalore–Hyderabad	$245.9\times$	$737.7\times$	2.39%	15.1%	0.2831

Two things stand out from these tables together. First, QBER and phase error are nearly identical across all four links, 2.39 to 2.46% and 15.1% respectively, because both

are set mainly by the misalignment and background parameters, which are treated as link-independent in this model; distance and elevation show up almost entirely through η_A, η_B and therefore through yield and final SKR, not through the error rates. Second, the spread in final key rate is large, a factor of 3.5 between the best link (Delhi-Nainital, 0.330 kbit/s) and the worst (Delhi-Mumbai, 0.095 kbit/s), and that spread is driven almost entirely by distance and elevation geometry rather than by which specific cities are paired.

5.3 Ground Site Ranking

Ranking the six ground sites purely on their literature-backed characteristics (lower aerosol loading, lower turbulence, higher altitude all favouring a site, all else equal) gives:

Nainital > Bangalore > Ahmedabad > Mumbai > Hyderabad > Delhi

This is a qualitative synthesis across the site table in Section 4.3, not a single computed index that QSAFire itself outputs. Table 5.3 summarises the basis for each site's position.

Table 5.3: Basis for the ground-site ranking.

#	Site	Basis for rank
1	Nainital	Lowest AOD (0.18) and highest altitude (1,951 m) of any site by a wide margin; only site with a turbulence multiplier below 1 ($\times 0.40$), corroborated indirectly by its real status as a working astronomical observatory site.
2	Bangalore	Second-lowest AOD (0.25); independently cross-checked by a global megacity study reporting Bangalore as the lowest-AOD megacity worldwide [9], though the same source flags the fastest proportional AOD increase of any megacity studied.
3	Ahmedabad	AOD 0.40 from real seasonal sun-photometer data, the best-instrumented of the newly added cities, with a full measured seasonal cycle rather than a single annual estimate [8].
4	Mumbai	AOD 0.45, sea-level altitude; aerosol data strong, but turbulence carried at the shared unenhanced baseline with only Indicative confidence, since a real Delhi C_n^2 dataset checked for reuse here turned out to use radio, not optical, refractivity and was unusable.
5	Hyderabad	AOD 0.55 from a 2009 baseline that the project's own more recent MODIS trend citation suggests understates current conditions (peak AOD 0.6–0.73 in recent years) [11].
6	Delhi	Heaviest aerosol burden of any site (AOD 0.78, rising to 0.91 in winter haze), confirmed by two independent field campaigns [6, 12]; the only site where the full annual simulation (Section 5.5) returns zero key volume in every season tested.

Delhi's and Hyderabad's positions are, if anything, conservative rather than pessimistic: both sites' own cited literature documents a worsening trend beyond the adopted baseline value, not a stable one.

5.4 Link Ranking

Table 5.4 ranks the four links by simulated secure key rate and gives the dominant geometric reason for each position.

Table 5.4: Links ranked by simulated secure key rate.

#	Link	SKR	Dist.	Peak elev.	Why
1	Delhi–Nainital	0.3300	234.5 km	75.7° / 76.4°	Shortest distance and Nainital's best-ranked site profile both push the same direction.
2	Mumbai–Ahmedabad	0.2911	440.5 km	86.3° / 87.0°	Near-zenith pass geometry, the highest of all four links, dominates despite medium distance.
3	Bangalore–Hyderabad	0.2831	500.6 km	78.2° / 78.7°	Bangalore's low aerosol partly offsets Hyderabad's dated, likely-understated AOD.
4	Delhi–Mumbai	0.0947	1,149.4 km	55.5° / 54.7°	Worst-ranked site paired with the longest distance and lowest elevation of any link; every disadvantage stacks on this one link.

The elevation/distance dependence driving this ranking is not an unconstrained model output: the underlying diffraction-loss formula is separately validated against Liao et al.'s real reported Micius figures (Section 5.6), so the ranking rests on a loss model anchored to a flown system's own reported performance. Delhi–Nainital's win despite including Delhi is a genuine result of the model, not an artefact of it: Nainital's short distance and highest partner-site elevation are large enough to outweigh Delhi's aerosol penalty at the Alice end.

5.5 Delhi–Mumbai: Seasonal and Annual Availability

Delhi–Mumbai is the only link carried through QSAFire's full seasonal, weather-gated annual model, since it is the primary case study. Table 5.5 gives the per-season key volume, relative to a fixed reference rate, for both stations.

Table 5.5: Per-pass key volume by season, relative to reference rate (corrected model).

Station	Season	$R_{\text{season}} / R_{\text{reference}}$
Delhi	Winter	0.000
Delhi	Summer	0.000
Delhi	Monsoon	0.000
Mumbai	Winter	0.111
Mumbai	Summer	0.021
Mumbai	Monsoon	0.037

Delhi does not close a link in any season tested; its combination of heavy year-round aerosol loading and the highest turbulence multiplier among the six sites keeps its seasonal availability at exactly zero throughout. Mumbai does close, though weakly, and best in winter. Table 5.6 converts this into annual key volume once weather-gating (the fraction of passes expected to be usably clear) is applied.

Table 5.6: Annual expected key volume, weather-gated (corrected model).

Station	Winter passes	Summer passes	Monsoon passes	Annual key bits
Delhi	54.4 clear	66.6 clear	21.4 clear	0
Mumbai	54.4 clear	66.6 clear	21.4 clear	1.88×10^5

Mumbai’s annual total of 1.88×10^5 key bits is made up of roughly 138,000 bits in winter, 31,500 in summer, and 17,900 in monsoon. This is also where the Priority 1 turbulence correction (Section 4.5) produces its one materially different conclusion, not just a different number: pre-correction, Mumbai’s winter scaling factor was 0.005, and its summer and monsoon factors were exactly zero, meaning the link only closed at all in winter, and weakly. Post-correction, winter rises to 0.111 (a 22-fold increase) and both summer and monsoon become nonzero for the first time. This happens because the corrected, smaller turbulence variance pulls Mumbai’s fading distribution away from the tail events that were killing the link under the uncorrected model. Delhi’s result is unaffected either way, since its aerosol and background-light burden alone are severe enough to close the link regardless of the turbulence correction.

Geometrically, before any weather-gating, 252 of 5,551 orbits per year (4.54%) produce a usable common-visibility pass between Delhi and Mumbai. Weather-gating reduces this to 175.5 expected fully-clear passes per year, concentrated in summer (mean joint availability 0.951) and winter (0.756), and heavily suppressed in monsoon (0.289), consistent with India’s monsoon cloud cover. A representative sensitivity sweep on this link ranks receiver aperture D_{Rx} as the most sensitive lever (62% range in outcome), minimum elevation angle second, and ground-level turbulence A_0 essentially irrelevant (0%

range), which is the same conclusion reached independently in Section 4.7: background light and geometry, not turbulence, are what actually bind this link.

5.6 Comparison Against Micius

Micius, launched in 2016, is the only satellite-QKD mission with published, peer-reviewed, in-orbit performance data, and is used throughout QSAFire as the primary real-world benchmark for orbit geometry, loss-budget validation, and final key-rate sanity-checking [5]. Table 5.7 consolidates every Micius cross-check made across Chapters 4 and this one into a single summary.

Table 5.7: Summary of every QSAFire-vs-Micius cross-check.

#	Check	QSAFire	Micius (real)	Status
1	Orbital inclination	97.40° @ 500 km	~500 km sun-sync	Pass
2	Slant range @ 85.7° elevation	501.31 km	507.0 km	Pass (−1.12%)
3	Slant range @ 25.0° elevation	1,031.94 km	1,034.7 km	Pass (−0.27%)
4	Per-station pass duration	279.5–282.5 s	~273 s/day	Pass (same order)
5	Ground-separation comparability	1,148 km (Delhi– Mumbai)	1,120 km (Delingha– Nanshan [19])	Pass (+2.5%)
6	Diffraction loss @ 1,200 km	25.25 dB	~22 dB	Same or- der
7	Atmosphere+turbulence loss	1.42–3.03 dB (shorter range)	3–8 dB @ 1,200 km	Pass (be- low range, expected)
8	Pointing loss (real Micius jitter)	−0.038 dB	<3 dB bud- get	Pass, well inside
9	Key rate, order of magnitude	26–75 bit/s	~100 bit/s (portable ground sta- tion)	Pass, same order
10	Key rate vs. Micius flagship	26–75 bit/s	1,200– 40,200 bit/s	Expected gap (dif- ferent hardware class)
11	Background suppression needed vs. demonstrated	209–407 × needed	> 10 ¹² × demon- strated [17]	Pass, >2,500 × margin

Checks 1 through 8 are geometry and loss-budget validations, and all pass to within a few percent or the same order of magnitude; these rest on orbital mechanics and Micius’s own published Methods figures, which is why they are treated as the strongest evidence in the project. Checks 9 through 11 concern the final key rate and are discussed in more

detail below, since a single Micius comparison number can be misleading if the wrong Micius figure is used.

5.6.1 The key rate comparison, three ways

Table 5.8 gives three different real benchmarks for the Delhi-Mumbai key rate, not one, and states which is the fair comparison rather than defaulting to the most flattering.

Table 5.8: Delhi-Mumbai key rate against three different real benchmarks.

Benchmark	Real reported value	QSAFire (Delhi–Mumbai)
Micius flagship, dedicated ground stations	40.2 kbit/s @ 530 km; 1.2 kbit/s @ 1,034.7 km [5]; ~9 kbit/s @ ~600 km, ~3 kbit/s @ ~1,000 km [20]	Delhi 26 bit/s, Mumbai 75 bit/s (~40–500× below)
Portable, non-ideal ground station (the fair comparison)	Sifted key rate at 1,000 km “can only reach the order of 100 bps” [21]	Delhi 26 bit/s, Mumbai 75 bit/s (same order)
Behera & Sinha, IAO Hanle (India-specific)	~2,080 bit/s sifted-key-equivalent order at 44.3 dB total loss [22]	Delhi 26 bit/s, Mumbai 75 bit/s (~28–80× below)

Micius’s flagship figure comes from purpose-built, fully-optimised 1 m telescopes at dedicated astronomical-quality sites, not the situation modelled here: Delhi and Mumbai are city-adjacent stations, with a realistic misalignment parameter ($e_a = 0.15$) and a 50%-efficiency detector rather than Micius’s flight-qualified hardware, and only modest, not fully-optimised, background suppression. The portable-ground-station study is explicitly about that same non-ideal regime, and QSAFire’s Delhi-Mumbai numbers land in the same order of magnitude as that real system, not the Micius flagship figure, which is why the portable-station comparison is treated as the meaningful one rather than the flagship number.

5.6.2 What Micius does and does not validate

Stated plainly, to avoid over-claiming from these comparisons: Micius validates geometry (orbit, elevation-range) and the shape and order-of-magnitude of the loss budget (diffraction, atmosphere-plus-turbulence, pointing); these are strong, multi-point, quantitative validations with hard-coded assertions in the notebook itself. Micius does *not* validate India-specific aerosol, turbulence, or background-light parameters; no part of the Micius comparison touches Delhi’s or Mumbai’s actual atmospheric or light-pollution

conditions, which come entirely from the separate, India-specific literature in Section 4.3. Micius is also a different protocol, a real entanglement-distribution and BB84-family system, not SNS-TF-QKD; the Delingha-Nanshan comparison (check 5 above) validates that a dual-downlink geometry at a similar baseline distance is real and demonstrated, not that Micius has flown this project’s specific protocol. Finally, the three non-primary links (Delhi-Nainital, Mumbai-Ahmedabad, Bangalore-Hyderabad) are not independently cross-checked against Micius the way Delhi-Mumbai is; Micius has never flown any of those ground-station pairs, so they inherit the same validated geometry and loss-budget machinery but do not have their own dedicated real-world data point.

5.7 Discussion

Taken together, the results in this chapter support three conclusions. First, QSAFire’s geometry and loss-budget machinery is well anchored to a real, flown satellite-QKD mission, to within a few percent on every quantity that can be checked directly against Micius’s own published Methods figures; this is the strongest claim the project can make, and it rests on orbital mechanics and reported hardware loss figures rather than on any of the India-specific proxy parameters. Second, the ranking of links is governed almost entirely by distance and elevation geometry rather than by which cities happen to be paired, to the point that Delhi, the worst-ranked individual site, still produces the best-performing link when paired with a short, high-elevation partner (Nainital). Third, and most importantly for the primary Delhi-Mumbai case study, the link’s performance is limited by night-sky background light and by Delhi’s aerosol and turbulence burden, not by the free-space turbulence physics that absorbed most of the mid-internship debugging effort; the implementation of spherical wave instead of plane wave changed the turbulence numbers substantially without changing this conclusion, which is itself a useful finding, since it means the project’s confidence in the headline results does not rest on getting the turbulence formula exactly right.

The main caveat that should travel with all of these results is the one stated in Section 4.10: no site in this study has a real, direct, optical ground-level C_n^2 measurement. Every result in this chapter that depends on a site’s turbulence multiplier, which is most of them, inherits that caveat, and closing it with a real measurement campaign is the single most valuable next step identified anywhere in this internship.

5.8 Chapter Summary

Across the four links studied, Delhi-Nainital performs best (0.330 kbit/s) and Delhi-Mumbai worst (0.095 kbit/s), a 3.5-fold spread driven by geometry rather than by ground-station identity. Delhi ranks worst of the six individual sites on aerosol and turbulence

grounds and is the only site that never closes a link across any season in the full annual model, while Mumbai's seasonal availability improved materially once the Priority 1 turbulence correction was applied. Every geometry and loss-budget mechanism checked against Micius's real published performance passes to within a few percent or the same order of magnitude, and the final Delhi-Mumbai key rate, while far below Micius's own flagship figures, lands in the same order of magnitude as real, non-ideal portable ground-station demonstrations, which is presented here as the fair comparison rather than the flattering one.

Chapter 6

Conclusion

6.1 Summary of Work Completed

This internship produced three independent technical deliverables under the shared broad theme of SNS-TF-QKD, reported separately in Chapters 2 through 5 because they were built separately, at different levels of physical implementation versus simulation, and share no code or data path with one another.

Task 1 built a three-station SNS-TF-QKD prototype in OptiSystem, with the protocol's window-selection, send/not-send, decoy-intensity, and phase-randomisation logic implemented as MATLAB co-simulation blocks, over 40 km fibre spans at a 10 Gbit/s symbol rate. The build produces synchronised, bit-index-keyed CSV logs from all three stations and was verified to run its protocol logic correctly end to end at a 1024-window sequence length. No secure-key-rate figure was claimed from this task on its own, since a run of that length is adequate to verify correctness but not to support a statistically defensible rate estimate.

Task 2 built a Python pipeline that ingests OptiSystem-style CSV exports and carries out the classical post-processing that Wang et al.'s SNS security analysis requires: synchronisation, basis separation, phase post-selection, decoy-state yield and phase-error estimation, Z-basis gain and QBER, and the Eq. (4) secret-key-rate calculation, with a Wilson confidence interval reported on every rate estimated from a finite count. Run against a 32,768-window CSV set, the pipeline executed end to end without error and returned a negative secret key rate, traced to detector clicks that did not appear statistically coupled to the underlying phase and send/not-send variables in that particular dataset, rather than to a fault in the pipeline itself.

Task 3, QSAFire, is an independent framework evaluating SNS-TF-QKD over a satellite-to-ground downlink, covering orbital geometry, atmospheric loss, turbulence, pointing loss, and a fully asymmetric, jointly-optimised secure-key-rate calculation, applied to six Indian ground sites and four candidate links. Its orbital solver and elevation-slant-range formula were validated directly against real published Micius data to within 1 to 3%, and a specific error in its turbulence model, use of the plane-wave rather than spherical-wave Rytov variance for an uplink geometry, was identified, initially mis-corrected twice on the basis of unverified assumptions, and then fixed properly using a citable literature ratio and confirmed by executing the corrected notebook end to end.

The corrected framework finds that Delhi-Nainital is the best-performing of the four links studied (0.330 kbit/s) and Delhi-Mumbai the worst (0.095 kbit/s), that background light rather than turbulence is the binding constraint on the Delhi-Mumbai link, and that this link's key rate, while roughly two orders of magnitude below Micius's flagship performance, lands in the same order of magnitude as real, non-ideal portable ground-station demonstrations reported in the literature.

6.2 Technical Contributions

Three contributions from this internship are worth stating explicitly rather than leaving implicit in the chapter narrative above.

First, the Task 1 OptiSystem build demonstrates that SNS-TF-QKD's protocol-level decision logic, which has no direct equivalent among OptiSystem's native components, can be implemented as a set of MATLAB co-simulation blocks operating on a shared, bit-index-synchronised logging convention across three independently-clocked stations. The specific implementation detail most likely to be reused, or to trip up a future reimplementer, is Bob's bit-value-convention flip relative to Alice's (Section 2.2), documented here because it is easy to miss and silently corrupts downstream sifting if missed.

Second, the Task 2 pipeline ties every computed quantity to a specific numbered equation in Wang et al. rather than to an ad hoc reimplementer, reports a confidence interval alongside every rate estimated from a finite count, and, when the input data produced a physically implausible result (a negative key rate), reported that result faithfully along with a specific, falsifiable diagnosis (detector clicks decoupled from the phase and SNS-category variables) rather than adjusting the output to look more reasonable.

Third, QSAFire's validation strategy, checking the orbital solver against two independent real missions (Landsat-8/9, Sentinel-2) before trusting it at the Micius-like altitude, checking the elevation-range formula against two independent published Micius data points, and checking the Rayleigh atmospheric baseline against a standard reference figure, means that the framework's core geometry and loss-budget machinery does not rest on any of the India-specific proxy parameters that carry the weakest evidence in the project. The turbulence correction documented in Section 4.5 is itself a contribution of process as much as of result: the two rejected attempts are recorded alongside the accepted fix specifically so that the eventual correction is understood as the output of a verify-then-trust process, not as a plausible-sounding claim accepted on the first attempt.

6.3 Limitations

The limitations of each task are stated in full at the end of their respective chapters (Sections 2.8, 3.9, and 4.10); the ones with the largest bearing on how these results should be

used are summarised here.

For Task 1, the 40 km fibre spans and 1024-window sequence length are laboratory-scale placeholders, not a demonstration of SNS-TF-QKD’s long-distance advantage, and no active phase-drift compensation stage is modelled at Charlie. For Task 2, the negative key rate obtained is a property of the specific CSV set exercised, most likely reflecting synthetic data that is structurally correct but not calibrated to a physically sensible channel, and the Eq. (3)/Eq. (4) cross-check did not numerically reconcile on this run, an open item rather than a resolved one. For Task 3, no site in the six-site study has a real, direct, optical ground-level C_n^2 measurement; every turbulence multiplier beyond the shared baseline is a proxy or an indicative estimate, and this is the single largest open item in the entire internship. QSAFire’s absolute background-light suppression multipliers are similarly indicative rather than measured, its $3\times$ suppression margin is a design convention rather than a derived quantity, and only the Delhi-Mumbai link carries a full seasonal, weather-gated annual model; the other three links are evaluated at a single representative pass.

6.4 Future Work

For Task 1, the natural next step is to run the prototype at a sequence length long enough, or repeated and concatenated across enough runs, to support a statistically defensible finite-key rate estimate, and to add an explicit phase-drift compensation stage before extending the fibre spans toward the distances where SNS-TF-QKD’s advantage over conventional QKD becomes apparent. For Task 2, the immediate priority is validating the pipeline against a CSV set from a properly noise-calibrated simulator run, in which detector clicks are physically coupled to the interference phase, to confirm that the same code returns a small, positive phase-error bound and a positive key rate as expected; reconciling the Eq. (3) and Eq. (4) discrepancy noted in Section 3.8 should be done at the same time. The modular package layout and automated PDF report generator described in Section ?? are natural follow-ups once that validation is complete. For Task 3, the highest-value next step by a clear margin is a real optical ground-level C_n^2 measurement campaign at one or more of the six sites, which would convert the framework’s largest proxy-based assumption into a directly measured one; a fully geometry-dependent slant-path turbulence weighting, rather than the constant 0.4065 conversion ratio used here, and a finite-key security treatment in place of the current asymptotic one, are the two next-most-valuable extensions to the physics model itself.

6.5 Closing Remarks

Across all three tasks, the results reported in this document are supported directly by executed code, real uploaded data, or literature citations checked before use, and every place where a result depends on an unverified or proxy assumption is flagged as such in the relevant chapter rather than left implicit. Task 1 and Task 2 establish that an SNS-TF-QKD protocol chain and its classical post-processing can be built and exercised correctly, without yet claiming a representative key-rate figure from either. Task 3 establishes that an SNS-TF-QKD satellite downlink to Indian ground stations is physically plausible under realistic loss conditions, with a computed performance consistent with real non-ideal ground-station demonstrations elsewhere in the world, and with its main open uncertainty, atmospheric turbulence at Indian sites specifically, identified precisely enough to be the clear target for any follow-on measurement work.

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Consolidated Parameter Reference

This appendix collects the global simulation parameters for all three tasks into single reference tables, for quick lookup without searching through each chapter. Full context, derivations, and citations for every value below are given in the chapter indicated in the final column.

Table 1: Task 1 (OptiSystem prototype) parameter reference.

Parameter	Value	Source
Bit rate	10 Gbit/s	Sec. 2.7
Sequence length	1024 bits/windows	Sec. 2.7
Samples per bit	32	Sec. 2.7
Laser frequency	193.1 THz (≈ 1552.52 nm)	Sec. 2.5
Laser output power	0 dBm	Sec. 2.5
Fibre type / length	SMF-28, 40 km $\times$ 4 spans	Sec. 2.5
Assumed fibre attenuation	0.2 dB/km (≈ 9 dB/span)	Sec. 2.5
Detector efficiency η_{det}	Configurable (30–80% range in literature)	Sec. 2.5

Table 2: Task 2 (Python pipeline) key result reference, 32,768-window CSV set.

Quantity	Value	Source
Q_X (X-basis gain)	0.1889	Sec. 3.5
E_X (X-basis error rate)	0.5588	Sec. 3.5
Q_Z (Z-basis gain)	0.1541	Sec. 3.7
$QBER = E^Z$	0.5810	Sec. 3.7
$s_0 = Y_0$ (vacuum yield)	0.1514	Sec. 3.6
$s_1 = Y_1$ (single-photon yield)	0.5082	Sec. 3.6
e_1^{ph} (phase-error bound)	0.6378	Sec. 3.6
R (secret key rate, Eq. 4, per window)	−0.1662	Sec. 3.8

Table 3: Task 3 (QSAFire) primary-link (Delhi–Mumbai) result reference.

Quantity	Value	Source
Orbital altitude / inclination	500 km / 97.40°	Sec. 4.2
Delhi–Mumbai distance	1,149.4 km	Sec. 5.2
Delhi–Mumbai peak elevation	55.5° / 54.7°	Sec. 5.2
Secure key rate (corrected)	0.0947 kbit/s	Sec. 5.2
Breakeven suppression (corrected)	854.2×	Sec. 5.2
Adopted suppression margin (3×	2,562.6×	Sec. 5.2
QBER / phase error	2.46% / 15.1%	Sec. 5.2
Annual key volume (Mumbai, corrected)	1.88×10^5 bits	Sec. 5.5
Annual key volume (Delhi)	0 (all seasons)	Sec. 5.5

Task 2 CSV Schema Reference

Table 4 lists the columns present in the Task 2 input files, as verified by the pipeline's own synchronisation check (Section 3.2).

Table 4: Required columns, Task 2 input CSVs.

File(s)	Required columns
alice_signal_record.csv,	Window_Index, Selected_Window, Send_Flag, Ran-
bob_signal_record.csv	dom_Number, Mu, Decision, Window_Type, De-
	coy_State, Decoy_Mu, Decoy_Random_Number, P0, P1,
	P2, Phase_Rad, Phase_Deg, Slice_Number, M
charlie_record_x.csv,	Window_Index, SPD1, SPD2, Detector, Event
charlie_record_z.csv	

Every row across every file is keyed by `Window_Index`, an integer window counter shared across all stations. `Event` takes one of four values (`D1_CLICK`, `D2_CLICK`, `DOUBLE_CLICK`, `NO_CLICK`); only the first two count as effective events in either basis.